MatchScore
Software Project Management Plan
Version 01.00

Revision History

Date	Version	Description	Author
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Software Project Management Plan

1. Introduction

1.1 Purpose of Project Management Plan

The purpose of this Project Management Plan is to define how the MatchScore project will be planned, executed, monitored, and closed over its 9-week development period. It documents the scope, schedule, resources, risks, and communication approach that will guide the project.
The intended audience of the MatchScore PMP is all project stakeholders, including the course instructor and the project developer.

2. Executive Summary of Project management plan

MatchScore is a web-based platform that allows job seekers to build a resume/portfolio, compare it against a job posting using a deterministic text-similarity algorithm (TF-IDF and cosine similarity), and receive AI-generated explanations and edit recommendations. This project will be developed by a single developer over a 9-week academic term, following an iterative build approach: document builder first, then the scoring engine, then the AI guidance layer.

2.1 Assumptions/Constraints

- The project will use a free-tier LLM API (e.g., Google Gemini or Groq); development is constrained by that API's rate limits.
- The developer has existing familiarity with React and Python/Flask, so minimal ramp-up time is budgeted.
- The MVP scope excludes a full portfolio-site builder and AI auto-edit; these are documented as stretch goals, given the compressed 9-week timeline.
- Development and testing will occur on the developer's personal machine; no dedicated staging/production server is budgeted.

3. Scope Management

The scope of MatchScore is limited to the resume builder (templates, WYSIWYG editor, PDF import/export), the job-fit scoring engine (TF-IDF and cosine similarity), and the AI guidance layer (explanation and recommendations). Portfolio-site building, AI auto-edit, and job-posting URL scraping are out of scope for this 9-week delivery and documented as future work. Any request to expand scope will be evaluated against the remaining timeline per the Change Control Management process (Section 3.3) before being accepted.

3.1 Work Breakdown Structure

WBS Item	Description
1.0 Planning	Define requirements, scope, and produce supporting documentation
1.1 Requirements Gathering	Define features, scope, and constraints
1.2 SRS Documentation	Software Requirements Specification
1.3 SDD Documentation	Software Design Document (architecture, diagrams, wireframes)
1.4 SPMP Documentation	Software Project Management Plan
2.0 Frontend Development	Build all user-facing screens for the application
2.1 Login Screen	User authentication UI
2.2 Registration Screen	Account creation UI
2.3 Template Selection Screen	Library, Manual Creation, and AI-Generated template tabs
2.4 Editor Screen	WYSIWYG, section-based resume/portfolio editor
2.5 PDF Import Review Screen	Review and correct extracted PDF content
2.6 Job Posting Input & Match Results Screen	Paste job posting, view score/terms/recommendations
2.7 Score History Dashboard Screen	Score trend chart and check-by-check history
3.0 Backend Development	Build the server-side services supporting the frontend
3.1 Auth Service	Registration, login, session management
3.2 Template Manager API	Serves library, manual, and AI-generated templates
3.3 PDF Export Engine	Renders finished document to downloadable PDF
3.4 PDF Import Engine	Extracts text from uploaded PDF
4.0 Scoring Engine	Compute a deterministic match score between resume and job posting
4.1 Preprocessor	Tokenization, stopword removal, stemming
4.2 TF-IDF Vectorizer	Converts text to weighted vectors
4.3 Cosine Similarity Calculator	Computes match score, matched/missing terms
5.0 AI Guidance Layer	Interpret the computed score and generate actionable feedback
5.1 Explanation Generator	Plain-language score explanation via LLM API
5.2 Recommendation Generator	Section-targeted edit suggestions via LLM API
6.0 Integration & Testing	Verify all modules work correctly together end-to-end
6.1 End-to-End Testing	Full builder → scoring → AI guidance flow
6.2 Defect Log Review	Track and resolve bugs found during testing
7.0 Deployment & Demo Prep	Prepare the final build and environment for submission

7.1 Hosting Setup	Frontend, backend, and database deployment
7.2 Demo Rehearsal	Walkthrough of full user flow before submission

3.2 Deployment Plan

MatchScore will be deployed as a single web application accessible via browser. The deployment approach involves: (1) hosting the frontend (React) on a static hosting service, (2) hosting the backend (Flask/Express) on a free-tier cloud platform, (3) provisioning the database (SQLite for development), and (4) configuring the AI API key as a secured environment variable. No data conversion or legacy system integration is required, since this is a new, standalone application.

3.3 Change Control Management

Any proposed change to MatchScore's scope, schedule, or feature set will be documented and evaluated for impact on the remaining timeline before being approved. Given the compressed 9-week schedule, any change request received after Week 5 will default to being deferred to future work unless it directly addresses a blocking defect. Example: if the free-tier AI API's rate limits prove insufficient during development, a change may be required to increase caching of AI-generated content — this would be evaluated against the remaining schedule before implementation.

4. Schedule/Time Management

The project schedule was baselined in Week 1 (current week) and will be reviewed weekly against actual progress. The developer is responsible for keeping the schedule updated and flagging any milestone at risk of slipping as early as possible, given the single-developer structure and the compressed 9-week timeline.

4.1 Milestones

The table below lists the milestones for this project, along with their estimated completion timeframe.

Risk	Likelihood	Impact	Mitigation
Login/Registration screen authentication logic (password hashing, session handling) takes longer than Week 2 allows	Medium	Medium	Use a well-documented auth library instead of building session handling from scratch; keep OAuth (Google) as stretch, not blocking
Editor Screen's section-based WYSIWYG structure is more complex to implement than estimated, delaying Week 3	Medium	High	Build with a proven rich-text library (e.g., TipTap) rather than a custom contentEditable implementation from scratch

PDF Import Review Screen's section-mapping (matching extracted text to Experience/Skills/Education) misclassifies content, delaying Week 3–4	Medium	Medium	Always show mapped sections for manual user correction (already in design); don't block on perfect auto-mapping
Scoring Engine (TF-IDF + Cosine Similarity) produces inconsistent or low-quality matches during Week 5 testing	Low	High	Test early with real sample resumes/postings, not placeholder text; validate against manually-scored examples
AI Guidance Layer hits free-tier API rate limits during Week 6 integration or later during demo	Medium	High	Cache explanation/recommendation results per resume+job pair; test demo flow ahead of time under real rate limits
Job Posting Input & Match Results Screen or Score History Dashboard Screen slip past Week 7 due to charting/UI complexity	Medium	Medium	Use a simple charting library (e.g., Chart.js) rather than building custom chart rendering
Insufficient time left for Week 8 integration testing if any prior screen/module slips	Medium	High	Treat Weeks 2–7 deadlines as hard checkpoints; cut AI Auto-Edit (stretch) immediately if behind schedule
Final deployment/demo environment (Week 9) behaves differently than local development environment	Low	High	Deploy to the chosen hosting platform by Week 8, not Week 9, so issues surface before the final demo

4.2 Project Schedule

https://app.clickup.com/90161096109/v/li/901617685949

4.2.1 Dependencies

- The AI Guidance Layer (Week 6) depends on the Scoring Engine (Week 5) being complete, since AI explanations require a computed score as input.
- Integration testing (Week 8) depends on all prior milestones being functionally complete.
- Final demo prep (Week 9) depends on integration testing passing without blocking defects.
- External dependency: continued availability of the chosen free-tier AI API within its published rate limits, throughout Weeks 5–9.

5. Cost/Budget Management

As a solo academic project, MatchScore has no monetary budget beyond optional hosting costs, expected to remain within free tiers (frontend hosting, backend hosting, and the AI API). No paid services are planned; if a paid tier becomes necessary, this will be raised as a change per Section 3.3.

6. Quality Management

Quality will be maintained through: (1) manual testing of each feature against its corresponding requirement in the SRS before it is considered complete, (2) verification that the match-score algorithm produces consistent, reproducible results for the same input pair, (3) review of AI-generated explanations and recommendations for relevance and adherence to the constraint that auto-edit never fabricates new claims, and (4) a defect log tracking any bugs found during testing until resolved and re-verified. Given the 9-week timeline, testing is scheduled as a dedicated activity in Week 8 rather than left until the final days before submission.

7. Human Resource Management

This project is developed by a single team member, who is responsible for all roles: requirements analysis, design, frontend development, backend development, AI integration, testing, and documentation.

8. Communications Management

As a solo project, most communication is between the developer and the course instructor. Progress updates will be shared according to the course's required check-in schedule.

8.1 Communication Matrix

Stakeholder	Messages	Vehicles	Frequency	Communicators	Feedback Mechanisms
Instructor	Progress updates, milestone deliverables	Email / LMS submission	Weekly (per milestone)	Developer	Instructor feedback/grading comments
Developer (self)	Task progress, blockers	Personal task tracker	Ongoing	Developer	Self-review against 9-week schedule

9. Risk Management

Risks will be identified as they arise and logged with a description, likelihood, impact, and mitigation plan. Given the compressed 9-week timeline, schedule risk is treated as the highest-priority category.

9.1 Risk Log

Risk	Likelihood	Impact	Mitigation
Document Builder (Weeks 2–4) slips, delaying everything after it	Medium	High	Treat Weeks 2–4 as the highest-priority window; cut portfolio-site stretch scope immediately if behind
Free-tier AI API rate limit hit	Medium	High	Cache AI responses per resume/job-posting pair;

during development or demo			pre-test demo flow ahead of time
PDF parsing fails on inconsistently formatted resumes	Medium	Medium	Support plain text input as a fallback; test with multiple sample PDFs early (Week 4)
Insufficient time left for integration testing	Medium	High	Keep Week 8 reserved exclusively for testing; do not let feature work bleed into it
AI Auto-Edit or portfolio-site scope creep delays MVP	Medium	Medium	Both are explicitly out of scope for the 9-week MVP (Section 3)

10. Issue Management

Issues (realized problems, as opposed to risks) will be logged with a description, date identified, and resolution status, and tracked until closed.

10.1 Issue Log

Maintained separately as a running log during development.
https://docs.google.com/spreadsheets/d/1tXh7F4Ehsk77Nhw6tfDdMeP_49KhYooVvMYIHLMGIak/edit?usp=sharing

11. Procurement Management

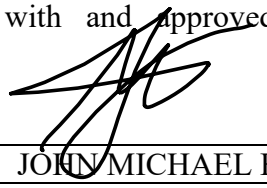
No procurement is required for this project. All tools used (React, Flask/Express, SQLite, free-tier AI API, hosting platforms) are free and self-provisioned by the developer.

12. Compliance Related Planning

MatchScore will avoid storing any sensitive personal data beyond what the user voluntarily enters into their resume, and will not share user resumes or job postings with third parties outside the AI API call required to generate explanations/recommendations. No additional regulatory compliance (e.g., HIPAA, FERPA) applies to this project's scope.

APPENDIX A: PROJECT MANAGEMENT PLAN APPROVAL

The undersigned acknowledge they have reviewed the ***MatchScore*** **Project Management Plan** and agree with the approach it presents. Changes to this **Project Management Plan** will be coordinated with and approved by the undersigned or their designated representatives.



Signature:		Date:	09/24/26
Print Name:	JOHN MICHAEL B. VILLAMOR		
Title:	AI Enthusiast		
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Signature:		Date:	
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Title:			
Role:			

APPENDIX B: REFERENCES

The following table summarizes the documents referenced in this document.

Document Name and Version	Description	Location
MatchScore SRS v01.00	Software Requirements Specification	MS Teams Shared Files
MatchScore SDD v01.00	Software Design Document	MS Teams Shared Files

APPENDIX C: KEY TERMS

The following table provides definitions for terms relevant to this document.

Term	Definition
TF-IDF	Term Frequency–Inverse Document Frequency, a text-weighting technique
Cosine Similarity	A measure of similarity between two text vectors
MVP	Minimum Viable Product
WBS	Work Breakdown Structure